Walk calibration of BTL modules in testbeam:

Via Synchronization of MCP scope data with FEBd

CMS

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Discussion draft; not approved for public release.

Purpose: provide a reliable time reference for BTL analyses

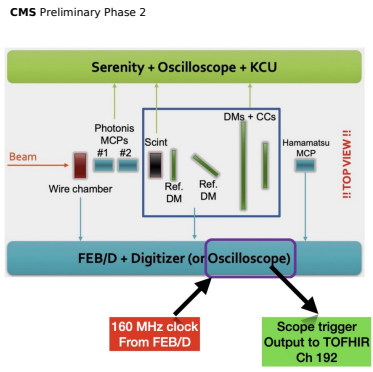
Several BTL test-beam analyses need a reliable particle-arrival time reference.

For a subset of the data, a digital oscilloscope recorded MCP-PMT signals in parallel with the BTL readout.

Approach: synchronize the two streams, use the MCP to calibrate upstream reference bars, then use their corrected times as the reference for downstream timing.

Final check: the time difference between downstream and reference bars before and after walk correction.

Parallel MCP and BTL streams are linked through FEBd

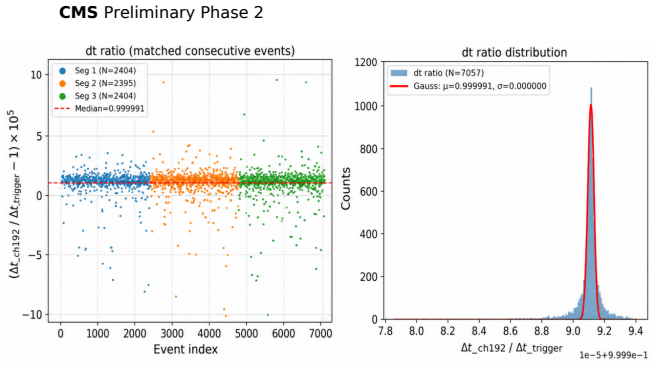


Two readout streams
The scope records the MCP-PMT pulse. FEBd records the BTL SiPM signals.

Synchronization link
The scope trigger also reaches TOFHIR channel 192 in FEBd. This link connects independently clocked records of the same particle.

September 2025 beam test. The highlighted lower path provides the reference-bar data. This FEBd-based synchronization is specific to the test-beam setup, not the final HL-LHC synchronization scheme.

Events are correlated using the intervals between triggers

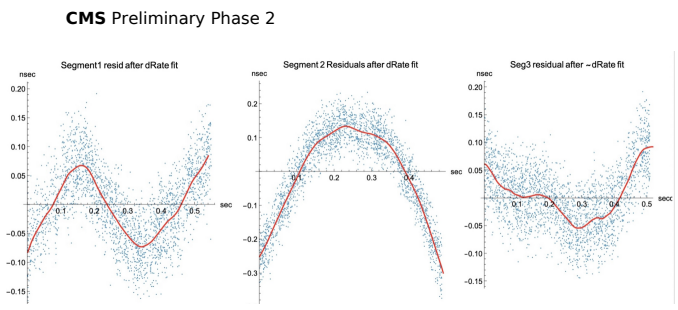


1. Compare consecutive trigger intervals in channel 192 and the scope.
2. Find consistent interval patterns.
3. Keep only unique event matches.

Consistent trigger patterns identify coincident events in the two streams.
Event matching alone does not remove relative clock drift.

The correlated intervals and their ratio test event matching, not detector resolution. GitHub: trigger matching.

The matched clocks show low-frequency relative drift



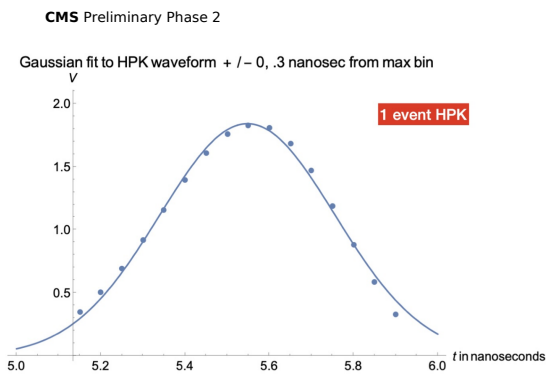
Correct the timing coordinate

1. Split data at clock resets.
2. Fit a line between matched trigger times in each segment.
3. Fit the remaining drift with LOWESS.

Add the drift to the line prediction to obtain $g(u)$, the expected TOFHIR trigger time.

Points show residuals before LOWESS; red curves describe smooth drift. Reject groups with pre-LOWESS residual scale $1.4826\,\mathrm{MAD} > 1$ ns. GitHub: clock reconstruction.

After synchronization, the MCP defines the calibration time



Fit the MCP waveform to estimate its pulse time.
Within the scope waveform, define

$$\delta_{\mathrm{MCP}} = t_{\mathrm{peak}} - t_{\mathrm{trigger}}.$$

For SiPM channel i, use

$$\Delta t_i = t_i - g(u) - \delta_{\mathrm{MCP}}.$$

The charge-dependent part of this difference is the walk.

The waveform and fitted template determine the MCP pulse time. u is the scope-trigger timestamp. A fixed cable/geometry offset is irrelevant to the shape; clock-rate conversion over the short waveform interval is neglected.

Measure the upstream channels' time walk versus charge

Measure the charge-dependent threshold delay relative to the synchronized MCP.

1. In run 4399, require exactly five TOFHIR hits and a unique MCP match: the target upstream pair, downstream channels 234 and 243, and channel 192.
2. Require waveform and clock quality, finite positive charge and slope-quality quantities. Keep charge at or above the spectrum's left half-maximum boundary.
3. Form 20 uniform $1/Q$ bins. Clip timing outliers at $5(1.4826\,\mathrm{MAD})$ from each bin's median; require at least five entries.
4. Fit the bin means with weights $1/\mathrm{SEM}^2$:

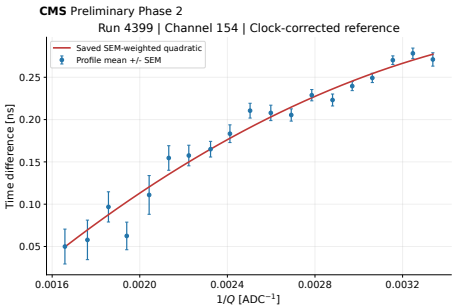
$$W_i(Q) = a_i + b_i/Q + c_i/Q^2.$$

Q is the measured ADC charge proxy, not energy in MeV. SEM denotes the standard error of the mean.

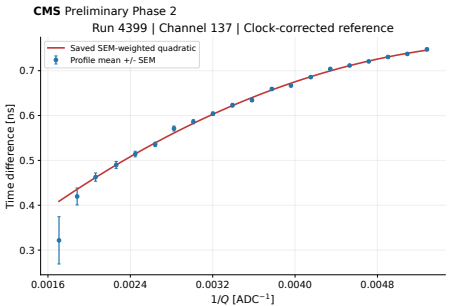
Reference pairs: bar 7 = channels 133 and 154; bar 8 = channels 137 and 150. GitHub: selected reference sample.

Walk-corrected upstream bars provide the time reference

Bar 7: channel 154



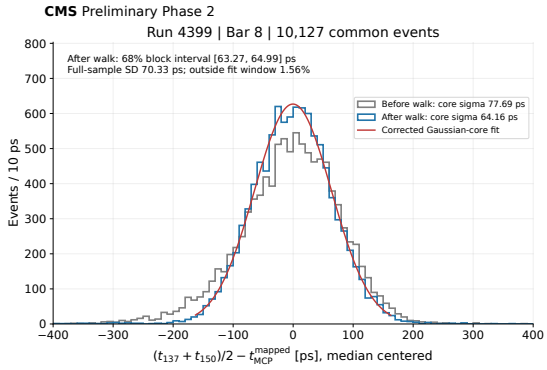
Bar 8: channel 137



Correct each end: $t_i^{\mathrm{corr}} = t_i - W_i(Q_i)$. Use the bar average: $t_{\mathrm{ref}} = (t_L^{\mathrm{corr}} + t_R^{\mathrm{corr}})/2$.

Run 4399 examples. Points show mean time differences with SEM; red curves are SEM-weighted quadratics in $1/Q$. Only populated charge support is shown. The $+26$ ns display offset does not change the walk shape. The full note also shows channels 133 and 150.

Check the calibrated reference bar against the MCP



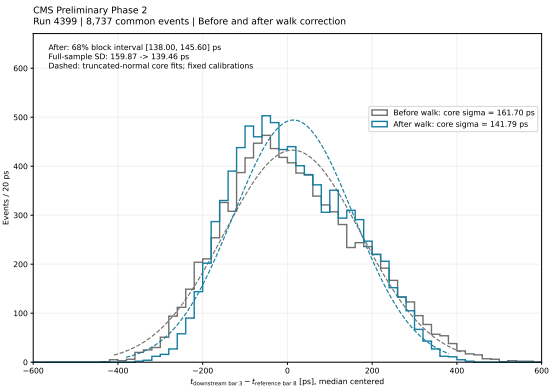
10127 common events
Subtract each channel's fitted walk, then form the bar-average time minus MCP time.
Core width

$$77.69 \longrightarrow 64.16 \text{ ps}$$

Corrected full-sample SD: 70.33 ps.
This is not the intrinsic bar resolution.

Fixed clock mapping; a truncated-normal core fit uses median $\pm 2.5(1.4826\,\mathrm{MAD})$. The corrected-width 16–84% interval is [63.27, 64.99] ps from 400 file/segment-block bootstraps with fixed calibrations.

Final check: downstream–reference-bar timing



8,737 common events
Bar 3 (234, 243) minus reference bar 8 (137, 150).
All four channels are walk corrected before averaging each bar's ends.

Core width

$$161.7 \longrightarrow 141.8 \text{ ps}$$

12.3% narrower. The MCP time cancels from this difference.

Same events and populated charge support; histograms median-centered for display. Same core-fit definition as the preceding slide; fixed-calibration 68% block interval: [138.0, 145.6] ps after correction. In-sample, asymmetric and fit-window dependent; not a single-bar resolution measurement.

References

[1] GitHub repository, `https://github.com/De-Cristo/LecroyTranslator4CMSTimingDetector`.

[2] Presentation by the Saclay group. `https://indico.cern.ch/event/1619706/`

[3] "Timing Performance of a Micro-Channel-Plate Photomultiplier Tube", `https://arxiv.org/pdf/1909.12604`

[4] CMS Collaboration, "A MIP Timing Detector for the CMS Phase-2 Upgrade". Technical Report CERN-LHCC-2019-003. CMS-TDR-020, CERN, Geneva, 2019 and "A High-Granularity Timing Detector for the ATLAS Phase-II Upgrade", CERN-LHCC-2020-007; ATLAS-TDR-031

[5] S. White, Nov. 2025 BTL TB Analysis mtg presentation, `https://indico.cern.ch/event/1614521/contributions/6803723/attachments/3178820/5922530/5vs10vs20GSa_s.pdf`

[6] W. R. Leo, "Techniques For Nuclear and Particle Physics Experiments", p.326, Springer-Verlag Berlin Heidelberg.

[7] S. White and A. Boletti, "Amplitude Walk in Fast Timing: The Role of Dual Thresholds" 2026 JINST V21 P02001.

[8] W. S. Cleveland, "Robust Locally Weighted Regression and Regression Scatterplots", J. Amer. Statist. Assoc. **74** (1979) 829.